Delta Lake Incremental Data Processing Assignment Report

1. Objective

The objective of this assignment is to demonstrate incremental data processing using Delta Lake in Databricks by loading data into a Delta table, cleaning it, processing incremental records with the MERGE operation, and validating the final output.

2. Software and Technologies Used

Databricks Serverless, Apache Spark (PySpark), Delta Lake, Python, and GitHub.

3. Implementation

The dataset was loaded into a Spark DataFrame and cleaned by removing null values and duplicate records. The cleaned data was stored as a managed Delta table. An incremental dataset containing updated and new employee records was created. The Delta Lake MERGE operation was then used to update matching records and insert new records efficiently.

4. Validation

The final Delta table was validated by checking the total row count and ensuring that no duplicate Employee IDs existed. The results confirmed successful incremental processing.

5. Results

The implementation successfully demonstrated Delta Lake's ability to perform efficient updates and inserts while maintaining data consistency and supporting ACID transactions.

6. Conclusion

This assignment provided hands-on experience with Delta Lake and PySpark in Databricks. It demonstrated a practical approach to incremental data processing commonly used in modern data engineering workflows.